Marco Domenico Buttiglione

PhD Candidate in XR & Real-Time Graphics • Game Developer

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PhD candidate at **Politecnico di Milano** (NECSTLab), now **Visiting Researcher at EVL**, University of Illinois Chicago. I build real-time XR systems, from **Gaussian Splatting** and **physics-based soft-body simulation** to multi-user VR, for medical training and research, and I direct the indie puzzle platformer **Paradox!**.

EXPERIENCE

Electronic Visualization Laboratory, UIC
Chicago, IL, USA

Visiting Researcher (host: Saeed Boorboor) — Sep 2026 – Present

- Real-time **3D Gaussian Splatting** for large head-tracked displays such as **CAVE2**, and volumetric and 4D capture for immersive training.

Politecnico di Milano · NECSTLab
Milan, Italy

PhD Researcher — Sep 2024 – Present

- Research on XR systems for training, mainly in the medical field.
- Built on real-time **3D Gaussian Splatting**, **XPBD** soft-body simulation and multi-user VR in Unity and OpenXR.

Teaching Assistant — Sep 2024 – Present

- *Computer Graphics* (Vulkan, MSc, in English) and *Laboratorio di Rappresentazione Digitale* (BSc, 3D CAD).

Research Fellow — May 2024 – Sep 2024

- **XPBD** soft-body framework for interactive anatomical Digital Twins in XR and surgical planning (ASME IDETC-CIE 2024, 2025).

Rewind Studios
Milan, Italy

Game Director & Developer — Oct 2022 – Present

- Directing a team of 7 on the puzzle platformer **Paradox!**: demo out on Steam, full game heading to release.
- Gamification consulting and client projects, including the 3D arcade game **CubeSnake**, co-developed with Kabum through release.

Wideverse
Bari, Italy

Frontend Web Developer — Jan 2021 – Sep 2021

- ReactJS CRM portal for the Italian Public Administration; Babylon.js 3D configurator for an outdoor blinds company.

EDUCATION

PhD, Information Technology
Politecnico di Milano — 2024 – now
Computer Science & Engineering area.
Advisor: Prof. Marco Gribaudo

MSc, Computer Science & Engineering
Politecnico di Milano — 2021 – 2024
Thesis: *Soft-Body Simulation for VR-Based Thoracoscopic Training*. 110/110 cum laude

BSc, Computer Science & Automation
Politecnico di Bari — 2017 – 2021
Thesis: *Cloud solutions for multimedia content distribution*. 110/110 cum laude

SKILLS

Engines & XR
Unity, Unreal Engine, OpenXR, Babylon.js

Graphics & Simulation
Vulkan, OpenGL, Gaussian Splatting, XPBD

Programming
C#, C++, Python, TypeScript, JavaScript

Web & ML
React, HTML/CSS, TensorFlow

LANGUAGES

Italian (native)
English (proficient)

SELECTED PUBLICATIONS

[1] **Cogito: A Dual-Role VR Platform for Standardized Cognitive Assessment with Behavioral Telemetry.** **M. D. Buttiglione**, A. Mancini, L. Ginestretti, M. Gribaudo. *ECMS 2026*.

[2] **Vendor Lock-In in VR Game Engines: A Structured Comparison of Unity, Unreal Engine, and Godot.** **M. D. Buttiglione**, F. Colabene, M. Gribaudo. *ECMS 2026*.

[3] **A Novel Pipeline for 3D Gaussian Splatting: Bridging Path Tracing and Real-Time Radiance Fields.** F. Costantini, **M. D. Buttiglione**, P. Piazzolla, M. Gribaudo. *ECMS 2026*.

[4] **Efficient Soft Body Simulation for Real-Time Applications: A GPU-Based XPBD Approach.** D. Alessi, **M. D. Buttiglione**, E. Lucania, P. Piazzolla, G. Colombo, M. Gribaudo. *ASME IDETC-CIE 2025*. BEST PAPER AWARD

[5] **Collaborative Virtual Reality Framework for Surgical Training and Simulation.** **M. D. Buttiglione**, F. Guerrera, P. Piazzolla, G. Colombo, E. Ruffini, M. Gribaudo. *ECMS 2025*.

[6] **Training Attention Skills in Individuals with Neurodevelopmental Disorders Using Virtual Reality and Eye-Tracking Technology.** A. Patti, F. Vona, A. Barberio, **M. D. Buttiglione**, I. Crusco, M. Mores, F. Garzotto. *HCII 2024, LNCS 14697*. BEST PAPER AWARD

[7] **Lung Operation Training in Low-Cost Virtual Reality Simulation Environments.** **M. D. Buttiglione**, F. Guerrera, P. Piazzolla, G. Colombo, M. Gribaudo. *ECMS 2024*.